

Airbnb Barcelona Price Prediction

A Multi-Model Approach to Structural Missingness

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Data Preprocessing & Consistency

- **Data Import:** Imported `listings.csv` and `reviews.csv`, mapping `listing_id` to `id` to fix a naming mismatch between the files.
- **Cleaning:** Dropped duplicate/unnecessary columns, filled missing data, and forced comments into strings.
- **Overlap Check:** Isolated only the reviews that successfully mapped back to the core `listings.csv` to ensure actionable data.

Guest Behaviour & Weighting

- **Review Frequencies:** Checked individual frequencies to see if we could add a “reviewer weight”.
- **Spoiler:** The distribution was heavily skewed (most left 1 review, but an active minority left 20+).
- **Fixing the Skew:** Applied a **log-scaled weight** to the exact amount of reviews a user left on a unique listing to fix the skewness.

NLP Model Selection & Formatting

- **Language Analysis:** Ran 5,000 comments through a language detection tool, confirming we must use a multilingual model.
- **The Model (mBERT):** Chose mBERT because it handles multiple languages and has a built-in scoring system trained on customer feedback.
- **Formatting Constraints:** Disabled model training for faster GPU performance and formatted text into uniform tensors (`padding = True`, `truncation = True` for comments over 512 tokens).

Scoring Math & Execution

- **Continuous Math:** We wanted a precise decimal number instead of just a 1-5 rating.
- **From Logits to Percentages:** The model calculates “softmax” percentages from “logits” (how strongly it feels about each rating).
- **Weighted Calculation:** It assigns a percentage (e.g., 5% for 1 star) and calculates:
 - $0.05 \times 1 \text{ (stars)} + 0.10 \times 2 \text{ (stars)} + \dots = \mathbf{3.70}$
- **Validation Check:** Fed the clean text list to the model and ran a quick check with top/least-scored comments to validate accuracy.

Aggregation & Temporal Features

- **Property Grouping:** Grouped scores by property to find the average median score, total reviews, and best/worst reviews.
- **Weighted Mean:** Extracted the weighted mean sentiment to get an overall score that considers experienced reviewers.
- **Review Consistency:** Calculated sentiment range (`max_score - min_score`) to tell if quality swings or is a steady experience.
- **Trend & Velocity:** Created a `sentiment_trend` feature (recent 12-month average minus older historical average) to see if reputation is going up or down.

Final Validation & Output

- **Data Splitting:** Split data into two sets to compare their effect on training:
 - **Unweighted Dataset:** Standard average (every reviewer is the same).
 - **Weighted Dataset:** Custom average (experienced reviewers considered more).
- **Pearson Correlation Check:** Compared the two datasets and got a correlation of **0.9954**. This told us the overall rating stayed consistent, but applied a subtle, targeted shift based on guest experience.
- **Cleanup:** Dropped temporary math columns and exported both files to compare results.